

Track 01 - LMS / LLL Website

Workshop & Learning Management Portal

Build a fully functional workshop and learning management portal where users can discover workshops, register or check out for them, access Google Meet or other meeting links, and manage their participation.

Platform Roles

- Admin - Creates and manages Organizer accounts; controls overall platform administration.
- Organizer - Creates and manages workshops, workshop activities and information; publishes workshop details for participants.
- Participant - Discovers workshops, registers or checks out, accesses meeting links, and tracks participation and attendance.

Key Requirements / Features

- Workshop listing and discovery
- Workshop registration and checkout
- Google Meet / online meeting links
- Attendance tracking
- Certificate generation
- Announcements
- New groups or communities
- Role-based access for Admin, Organizer and Participant
- Organizer management by Admin

Certificate Requirement

- 90% attendance is mandatory for certificate eligibility.
- The system must track attendance and verify the required percentage before generating a certificate.

Core Flow

Admin creates Organizer account -> Organizer creates Workshop -> Participant registers / checks out -> Participant attends -> Attendance is recorded -> 90% attendance verified -> Certificate generated

Objective

Create a working workshop and learning management platform, not merely a static website.

Track 02 - OCR

Digitizing Tamil Knowledge Resources

Build an OCR-based solution that converts Tamil content from scanned or digital sources into usable digital Tamil text. The solution should make Tamil knowledge resources digitally accessible and usable.

Target Sources

- PDFs
- Tamil textbooks
- Books
- Manuscripts (suvadi) / Epigraphs
- Other Tamil resources
- Classical Tamil text resources

Core Requirement

- The system should take Tamil content available as scanned pages, images or other digital source formats and process it into usable digital Tamil text.

Possible Capabilities

- Tamil text extraction
- Page-wise processing
- Text correction
- Extracted-text editing
- Searchable digital text
- Export/download of extracted text

Core Flow

PDF / Book / Manuscripts (suvadi) / Epigraph / Image -> OCR Processing -> Tamil Digital Text

Objective

Help digitize Tamil knowledge resources and make their content available in a usable digital format.

Track 03 - WordNet

Explore Tamil Words Across Literature and Context

Build a WordNet-style lexical exploration system that helps users understand a Tamil word by discovering its occurrences, meanings and usage across different literary resources.

Example Experience

- A user reading a Thirukkural clicks a Tamil word.
- The system identifies the word and finds occurrences across available literary resources.
- It displays relevant examples from poems, stories, songs and other Tamil literary resources.
- It provides the meaning of the word according to its context - not just one generic dictionary meaning.

Possible Features

- Clickable Tamil words
- Word search
- Word occurrences across texts
- Contextual meanings
- Literary references
- Related words
- Source information

Core Flow

Read Tamil Text -> Click a Word -> Find the Word Across Literature -> View Relevant Occurrences -> Understand Its Meaning in Context

Objective

Create an intelligent way for users to explore Tamil vocabulary through actual literary usage.

Track 04 - Learning Portal

Learn Tamil from the Beginning

Build a structured Tamil learning portal that allows a learner to start from the fundamentals and progressively develop the ability to read and write Tamil.

Learning Progression

• Tamil Basics -> Tamil Letters -> Writing Practice -> Words -> Sentences -> Reading -> Understanding Tamil Texts -> Reading Tamil Books

Possible Features

- Beginner-friendly lessons
- Tamil alphabet learning
- Tamil writing practice
- Vocabulary learning
- Reading exercises
- Quizzes
- Progress tracking
- Learning milestones
- Achievement / progress indicators

Experience Requirement

- The platform should provide a milestone-based learning journey rather than disconnected lessons.
- The learner's progress should be clear and the path should remain structured from beginning to advanced reading.

Core Flow

Learn -> Practice -> Take Quiz -> Complete Milestone -> Progress to Next Stage

Objective

Help learners progress from Tamil fundamentals to confidently reading and engaging with Tamil books.

Track 05 - Chatbot

Universal Knowledge Assistant & AI-Powered Knowledge Discovery

Build an intelligent, multilingual, source-grounded Universal Knowledge Assistant capable of understanding natural-language queries and dynamically discovering the correct information from relevant sources within a digital platform.

Supported Knowledge Sources

- Webpages
- Databases
- Books
- PDFs
- Documents
- CMS content
- Digital archives
- Multilingual resources
- E-books
- Chapters and sections
- Online reading links
- Downloads

Core Requirement

- The chatbot must not depend on predefined questions, fixed intent mappings, hardcoded keywords, or page-specific search rules.
- It should dynamically understand the query, determine where the required information exists, retrieve relevant content and provide a contextual answer.

Book Search Example

- For a book-related question, search beyond metadata and consider chapters/sections, actual digital content, e-book/PDF, online reading links, downloads and associated resources.

Multilingual Understanding

- Understand English, Tamil, Tenglish, transliteration, mixed-language queries, vague queries and natural conversational queries.
- Follow-ups such as 'Who wrote this?', 'What does it say about this?' and 'Can I read it online?' should be resolved using conversation context.

Core Flow

User asks a natural-language question -> Understand language and context -> Identify required information -> Discover relevant platform sources -> Retrieve verified content -> Rank sources -> Generate contextual answer -> Provide authentic source/resource link

Objective

Transform conventional search/chat into accurate, contextual, multilingual and source-verifiable knowledge discovery.

Context Awareness

- Preserve conversation context
- Understand follow-up questions
- Connect current queries with previous messages
- Resolve what the user is referring to
- Retrieve information relevant to the ongoing conversation

Dynamic Source Discovery

- Determine what the user is asking -> locate the relevant source -> retrieve verified content -> rank relevant sources -> generate the answer.

Source-Grounded Answers & No Fabrication

- Answers must be based on verified platform content and provide authentic source/resource links whenever available.
- Never invent, substitute unrelated information, rely on keyword-only matches, or present unverified information as factual.
- If information cannot be verified from available platform content, clearly state that it is unavailable.

Dynamic Knowledge Updates

- New webpages, books, PDFs, documents and digital resources should become searchable without developers manually creating individual question-answer rules.

Track 06 - Open Innovation

Build Something Useful for Classical Tamil

This is an open-ended innovation track. Participants can build any useful technology solution related to the Central Institute of Classical Tamil (CICT) and its areas of work.

Open Innovation Scope

- Unlike the other tracks, participants are not restricted to a predefined problem type.
- Teams may identify a meaningful problem and develop an original technology-based solution.

Potential Areas

- Classical Tamil resources
- Digital preservation
- Tamil literature
- Research support
- Tamil education
- Digital libraries
- Search and discovery
- Language technology
- Accessibility
- Knowledge management
- Cultural heritage
- AI / NLP applications

Focus

- The listed areas are possible directions, not restrictions.
- Solutions should emphasize originality, usefulness and practical application.

Core Flow

Identify a meaningful problem related to CICT / Classical Tamil -> Design and build a useful technology solution -> Create meaningful impact

Objective

Encourage original, practical technology solutions that create meaningful value for Classical Tamil and CICT-related work.